

MEE 342 Project Phase 3

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Fabrication Details:

The final Prototype was fabricated using PLA filament on a Bambuu Labs P1S. The model was reprinted only 1 time due to issues with the assembly rather than actual printing issues. The biggest issue during assembly was mistakenly gluing the shaft to the ring gear, causing it to not be able to rotate freely. This made it so the assembly was unable to carry out its action properly.

Assembly Procedures and Details:

All of our prototype parts were printed as separate pieces fastened together using end caps and a central support for the main spider and side gears. It was quite challenging to figure out a way to have ample support while also allowing the shafts and gears to be able to rotate freely. Another main issue was that we were unable to create a strong base for the model without making it extremely difficult to assemble. This meant we had to use a lot of external support when carrying out the action of the prototype.

Test Procedures:

To test the prototype, we assembled it and then cranked the shafts by hand to simulate the action of the differential in a real automobile. We first tested straight line driving by spinning the pinion gear. The pinion gear drove the spider and side gears with them all spinning at the same speed. To simulate a left turn we held the right side of the shaft stationary while spinning the opposite end. This caused 3 of the gears to rotate and the right one to stay stationary. The same process was done to simulate a right turn. All tests showed that the model was successful and carried out the right functions.

Failures/Mistakes:

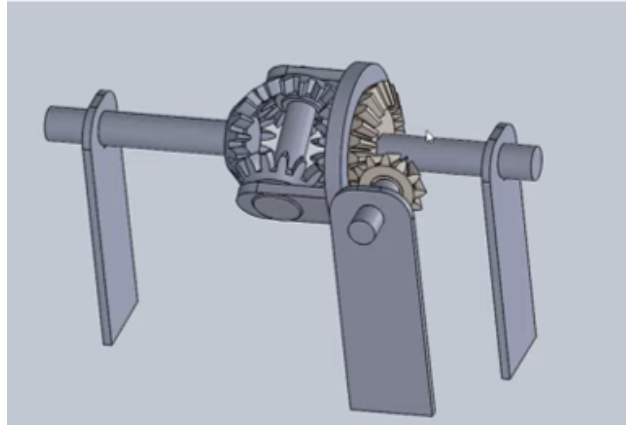
During the design, we had to model a component to hold our spider gears in place. To reduce the effect of the shaft's friction on the spider gears' rotation, we reduced the outer diameter of the shaft, but this led to the spider gears being able to shift while assembled, causing misalignment. Another mistake made was the lack of proper support for the model. We were able to demonstrate it worked, but because of all the torque, the small supports we made would just slip out, and the model would fall apart.

Our final mistake was in the assembly of the model; instead of modeling the ring gear and spider gear holder as one piece and printing them as one, we had to glue them together. When gluing, a mistake was made, and a shaft was adhered to the gears when it shouldn't have been, so a reprint was needed.

Possible Design Changes:

One possible design change would be our choice of material for analysis; we chose a random, non-structural steel material when we could have done more research and selected a material like 4130 steel. A final design change would have been the use of bearings on most of the gears, with the greatest effect being on the spider gears, and the previously mentioned support to hold them in place wouldn't have had misalignment issues by having bearings fill in the gaps while allowing full rotation

Modeled Assembly:



Prototyped Assembly:



Final Assembly:

